LABORATORY RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Experiment No.: 2
Student Name: S Satya Jayavanth
Roll No.: A24126552116
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1. TITLE

Student Profile Manager using JavaScript Class and DOM

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2. AIM

To develop a **Student Profile Manager** using JavaScript classes and DOM manipulation to accept student details from the user and dynamically display the student profile on a webpage.

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3. OBJECTIVE

The objectives of this experiment are:

- To understand the concept of JavaScript classes and objects.
- To create a Student class with appropriate properties.
- To use a constructor to initialize student details.
- To accept student information through HTML input fields.
- To retrieve input values using DOM methods.
- To create a Student object using user-provided values.
- To dynamically generate and display HTML content using JavaScript.
- To apply CSS styling to the dynamically generated student profile.
- To understand the interaction between HTML, CSS and JavaScript.

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4. THEORY

A **class** in JavaScript is a blueprint for creating objects. It defines the properties and methods that an object can have. The constructor() method is used to initialize the properties of an object when it is created.
In this experiment, a Student class is created with four properties: **name, roll number, department and CGPA**. The values for these properties are obtained from input fields provided on the webpage.
The **Document Object Model (DOM)** represents an HTML document as a collection of objects. JavaScript can use the DOM to access and modify HTML elements dynamically. The document.getElementById() method is used to select elements based on their IDs.
After retrieving the values from the input fields, a new object of the Student class is created. The student's details are then dynamically inserted into the profile <div> using the innerHTML property.
CSS is used to style the input fields, button, container and dynamically generated student profile.

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5. ALGORITHM / PROCEDURE

- Create an HTML document with the required structure.
- Add input fields for Name, Roll Number, Department and CGPA.
- Add a button to create the student profile.
- Create an empty <div> with the ID profile to display the generated profile.
- Create a JavaScript class named Student.
- Define a constructor inside the class.
- Initialize name, rollNumber, department and cgpa properties using the constructor.
- Create a function named createProfile().
- Use document.getElementById() to retrieve the values entered by the user.
- Create a new object of the Student class using the retrieved values.
- Select the profile div using its ID.
- Generate the student profile using a template literal.
- Insert the generated content into the profile div using innerHTML.
- Apply CSS styling to the webpage and profile.
- Open the webpage in a browser.
- Enter the student details and click **Create Student Profile**.
- Verify that the student information is displayed dynamically.

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6. PROGRAM

Task2.html

```
<!DOCTYPE html>
<html lang="en">
```

```html
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Profile</title>
  <link rel="stylesheet" href="Task22.css">

</head>

<body>

  <div class="container">

    <h1>Student Profile</h1>

    <input type="text" id="name" placeholder="Enter Name">
    <input type="text" id="roll" placeholder="Enter Roll Number">
    <input type="text" id="department" placeholder="Enter Department">
    <input type="number" id="cgpa" placeholder="Enter CGPA" step="0.01">

    <br>

    <button onclick="createProfile()">Create Student Profile</button>

    <div id="profile"></div>

  </div>

  <script src="Task23.js"></script>

</body>

</html>
```

Task2.js

```javascript
// Student class
class Student {

  constructor(name, rollNumber, department, cgpa) {
    this.name = name;
    this.rollNumber = rollNumber;
    this.department = department;
    this.cgpa = cgpa;
  }
```

```
}

// Function to create student profile
function createProfile() {

    // Get values from input fields
    let name = document.getElementById("name").value;
    let rollNumber = document.getElementById("roll").value;
    let department = document.getElementById("department").value;
    let cgpa = document.getElementById("cgpa").value;

    // Create Student object
    let student = new Student(
        name,
        rollNumber,
        department,
        cgpa
    );

    // Select profile div
    let profile = document.getElementById("profile");

    // Dynamically create HTML content
    profile.innerHTML = `
            <h2>Student Profile</h2>
            <p><b>Name:</b> ${student.name}</p>
            <p><b>Roll No:</b> ${student.rollNumber}</p>
            <p><b>Department:</b> ${student.department}</p>
            <p><b>CGPA:</b> ${student.cgpa}</p>
        `;
}
```

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7. CODE EXPLANATION

HTML Code
<input type="text" id="name">
Creates an input field for entering the student's name.
<input type="text" id="roll">
Creates an input field for entering the student's roll number.

<input type="text" id="department">
Creates an input field for entering the student's department.
<input type="number" id="cgpa">
Creates a numerical input field for entering the student's CGPA.
<button onclick="createProfile()">
Creates a button that calls the createProfile() function when clicked.
<div id="profile"></div>
Provides an empty area where the student profile is dynamically displayed.
JavaScript Code
class Student
Defines a class named Student.
constructor(name, rollNumber, department, cgpa)
Defines the constructor that receives the student's details when a Student object is created.
this.name = name;
Stores the student's name in the object.
this.rollNumber = rollNumber;
Stores the student's roll number.
this.department = department;
Stores the student's department.
this.cgpa = cgpa;
Stores the student's CGPA.
function createProfile()
Defines the function responsible for creating and displaying the student profile.
document.getElementById("name").value
Retrieves the name entered in the input field.
document.getElementById("roll").value
Retrieves the roll number entered by the user.
document.getElementById("department").value
Retrieves the department entered by the user.
document.getElementById("cgpa").value
Retrieves the CGPA entered by the user.
new Student(...)
Creates a new object of the Student class using the entered values.
document.getElementById("profile")
Selects the profile <div> from the webpage.
profile.innerHTML = ...
Dynamically inserts the student profile content into the webpage.
Template literal ${student.name}
Displays the corresponding property of the Student object inside the generated HTML.

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8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file	Student Profile page should be displayed	Page displayed successfully	Pass
TC-02	Enter student name	Name should be accepted	Name accepted	Pass
TC-03	Enter roll number	Roll number should be accepted	Roll number accepted	Pass
TC-04	Enter department	Department should be accepted	Department accepted	Pass
TC-05	Enter CGPA	CGPA should be accepted	CGPA accepted	Pass
TC-06	Click Create Student Profile	Student object should be created	Object created successfully	Pass
TC-07	Verify profile display	Student details should appear dynamically	Profile displayed successfully	Pass
TC-08	Enter different student details	New details should be displayed	Updated profile displayed	Pass

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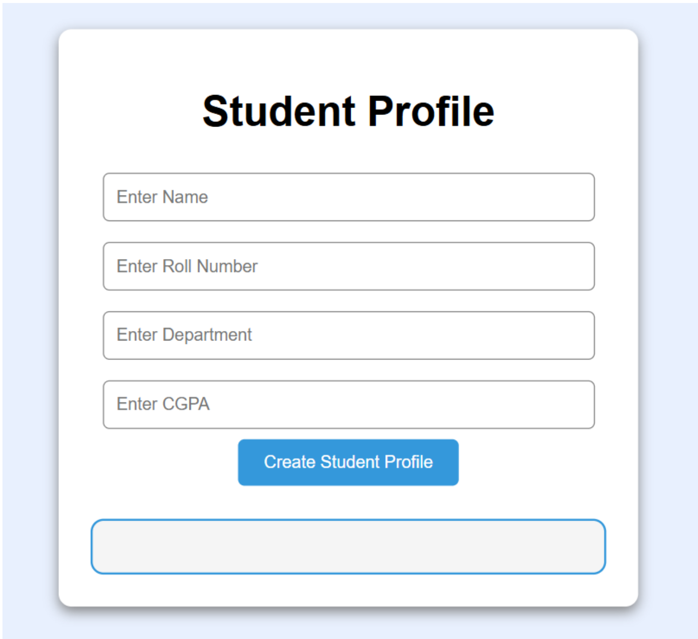
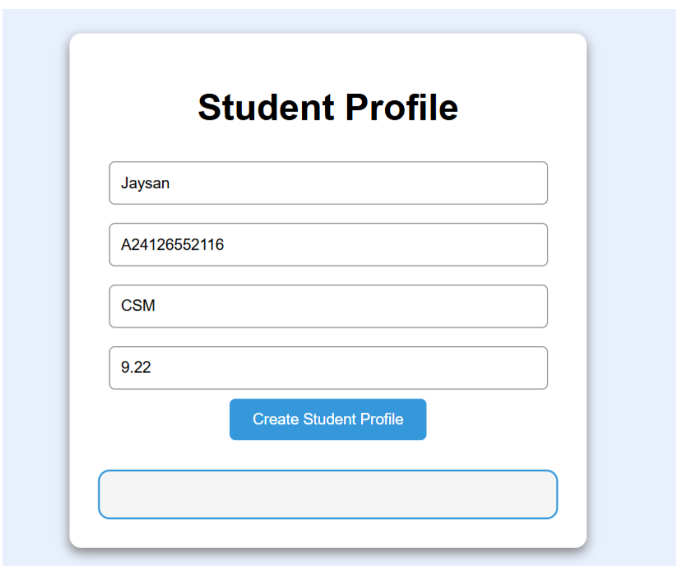
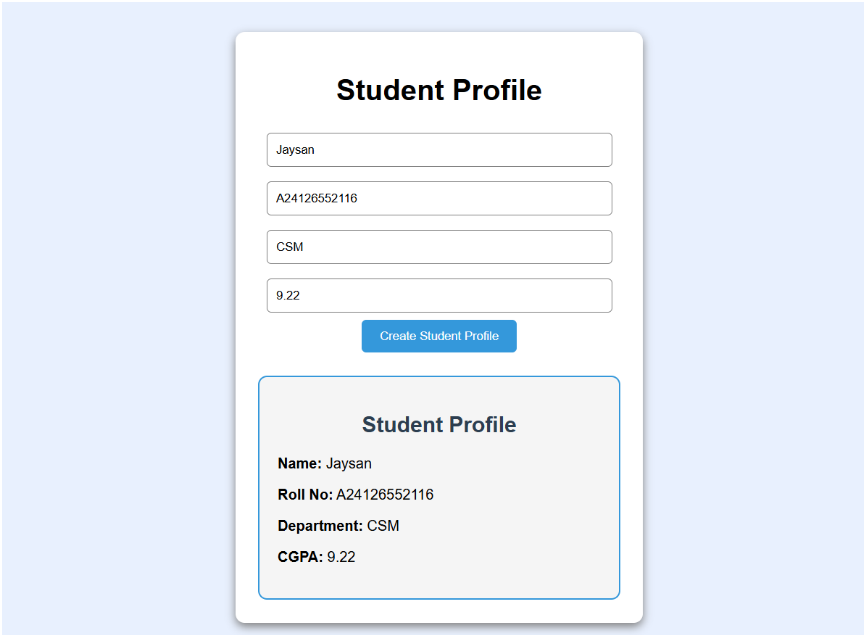
9. OUTPUT

Webpage


Student Profile
Enter Name
Enter Roll Number
Enter Department
Enter CGPA
Create Student Profile

Working


Student Profile
Jaysan
A24126552116
CSM
9.22
Create Student Profile



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10. RESULT

Thus, the **Student Profile Manager** was successfully developed using a JavaScript class and DOM manipulation. The program successfully accepts student details from the user, creates a Student object using the class constructor, and dynamically displays the student's **name, roll number, department and CGPA** on the webpage with CSS styling.